

1.0 Patient Registration

READ Registration Information from Patient or Corporate Client

IF Patient is a new record THEN

VALIDATE Registration Information

GENERATE new Patient ID

WRITE Patient Details to D1 Patient Records

ELSE

RETRIEVE existing Patient Details from D1 Patient Records

UPDATE specific Patient Details

WRITE updated data to D1 Patient Records

ENDIF

RETURN Registration Status to Patient

2.0 Questionnaire & Consent

READ Patient ID

PROMPT Patient with Questionnaire and Consent Form

RECEIVE Questionnaire Responses and Consent Authorization

VERIFY completion of required fields

WRITE response data and authorization to D2 Questionnaire Responses

RETURN confirmation to Patient

3.0 Appointment & Queue Management

READ Appointment Request from Patient

IF Request is for a Scheduled Appointment THEN

CHECK availability in D3 Appointment & Queue

ASSIGN schedule slot

WRITE appointment details to D3 Appointment & Queue

ELSE IF Request is for Walk-in THEN

GENERATE Queue Number

WRITE queue assignment to D3 Appointment & Queue

ENDIF

OUTPUT Queue Number or Appointment Confirmation to Patient

4.0 Clinical Assessment & Diagnostics (Level 2 Breakdown)

4.1 Patient Check-in

READ Patient ID and Appointment Data from D3 Appointment & Queue

UPDATE appointment status to "Arrived/Checked-in"

NOTIFY Nurse/Cashier of patient arrival

4.2 Nurse Assessment

READ Patient ID

INPUT Symptoms and Vitals from Nurse Assessment

WRITE assessment data to D4 Clinical Records

ROUTE Patient to Doctor or Laboratory

4.3 Laboratory Request

IF Doctor requests laboratory tests THEN

GENERATE Laboratory Request

TRANSMIT Laboratory Request to Laboratory Staff

UPDATE D4 Clinical Records with pending lab order

ENDIF

4.4 Laboratory Result Encoding

READ Laboratory Results from Laboratory Staff

FORMAT result data

WRITE Laboratory Results to D4 Clinical Records

NOTIFY Doctor and AI Assistant Services of new results

4.5 Clinical Intelligence Assistant

READ updated data from D4 Clinical Records

ANALYZE data against D5 AI Knowledge Base

GENERATE Clinical Insights and AI Medical Summary

WRITE Insights to D4 Clinical Records

4.6 AI Chatbot / Agent

RECEIVE chat query from Doctor, Nurse, or Patient

RETRIEVE context from D4 Clinical Records and D5 AI Knowledge Base

GENERATE Clinical Guidance response

OUTPUT response to respective user

4.7 Final Consultation & EHR Update

READ Clinical Insights, Laboratory Results, and Nurse Assessment from D4 Clinical Records

INPUT Doctor's Diagnosis and Prescription

UPDATE D4 Clinical Records with final encounter details

WRITE finalized summary to D1 Patient Records

5.0 AI Assistant Services

READ new clinical summaries and diagnosis data from D4 Clinical Records

PROCESS data through AI learning models

EXTRACT new medical patterns and anonymized insights

WRITE updated insights to D5 AI Knowledge Base

6.0 Healthcare Information Exchange

RECEIVE Corporate Medical Request or External HIE Query

READ Consent Authorization from D2 Questionnaire Responses

IF Consent is Valid and Active THEN

RETRIEVE requested data from D6 Shared Healthcare Records

FORMAT data into standardized exchange protocol

TRANSMIT Patient Medical Record securely

ELSE

REJECT request and return "Unauthorized Access" error

ENDIF

7.0 Billing & Payment

READ Services, Laboratory Tests, and Prescriptions from D4 Clinical Records

COMPUTE Total Bill amount

GENERATE Billing Information

OUTPUT Billing Information to Nurse/Cashier and Patient

RECEIVE Payment Confirmation

WRITE transaction details to D7 Billing Records

GENERATE and OUTPUT Receipt to Patient or Corporate Client